5.1 Functions and their Graphs

Some basic concepts

5.1.1 Quantity

Anything which can be performed on basic mathematical operations like addition, subtraction, multiplication and division is called a quantity.

5.1.2 Constant

A quantity which retains the same value throughout a mathematical investigation is called a constant.

Basically constant quantities are of two types

(i) **Absolute constants** are those which do not change their values in any mathematical investigation. In other words, they are fixed for ever.

Examples: 3, $\sqrt{3}$, π, ...

(ii) **Arbitrary constants** are those which retain the same value throughout a problem, but we may assign different values to get different solutions. The arbitrary constants are usually denoted by the letters $a, b, c, \ldots$ **Example:** In an equation $y=mx+4$, m is called arbitrary constant.

5.1.3 Variable

A variable is a quantity which can assume different values in a particular problem. Variables are generally denoted by the letters $x, y, z, \ldots$

Example: In an equation of the straight line $\frac{x}{a}+\frac{y}{b}=1$,

x and y are variables because they assumes the co-ordinates of a moving point in a straight line and thus changes its value from point to point. a and b are intercept values on the axes which are arbitrary.

There are two kinds of variables

(i) A variable is said to be an **independent variable** when it can have any arbitrary value.

(ii) A variable is said to be a **dependent variable** when its value depend on the value assumed by some other variable.

Example: In the equation
$y = 5x^2 - 2x + 3$,

"x" is the independent variable,

"y" is the dependent variable and

"3" is the constant.

5.1.4 Intervals

The real numbers can be represented geometrically as points on a number line called real line. The symbol R denotes either the real number system or the real line. A subset of the real line is an interval. It contains atleast two numbers and all the real numbers lying between them.

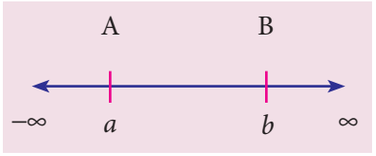


Fig. 5.1

(i) Open interval

The set $\{x : a < x < b\}$ is an open interval, denoted by (a, b).

In this interval, the boundary points a and b are not included.

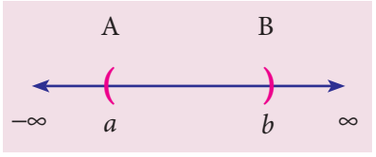


Fig. 5.2

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